

Development of a 3D-printable Virtual Impactor for Fukushima Daiichi decommissioning issues

End-of-studies Engineering Internship Defense
5A EMME MFE - Nuclear Engineering Track



Pierre ROGER | September 14, 2026

Introduction: Fukushima Daiichi Nuclear Power Plant Accident

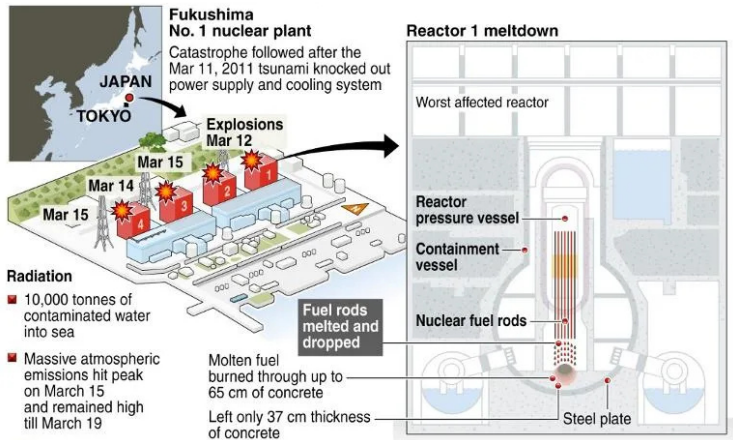
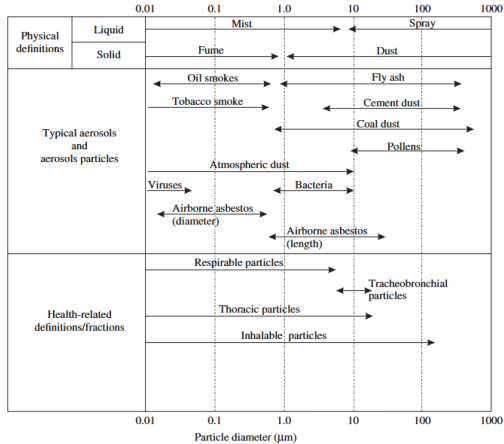


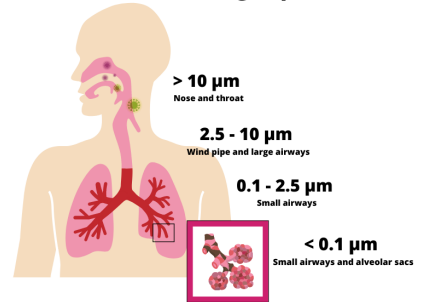
Figure: The Fukushima Nuclear Disaster (TEPCO/Atmospheric Chemistry and Physics).

⇒ **Radioactive aerosols were created by FDNPP accident or will be created through decommissioning.**

Introduction: Aerosols



Particle Size & Lung Deposition



(<https://deohs.washington.edu/healthy-air-healthy-schools-technical-informationpm>)

Figure: Particle size range for aerosols (Colbeck et al., 2014).

Introduction: Radioactive aerosols concern

D_{eff} : Effective dose received through interaction with radioactive isotope

$$D_{\text{eff}} = \underbrace{e(50)}_{\text{Effective dose coefficient (Sv}\cdot\text{Bq}^{-1})} \times \underbrace{a_s \times m_{\text{particles}}}_{\text{Activity (Bq)}}$$

$$\Rightarrow D_{\text{eff}} = f(d_{\text{ae}})$$

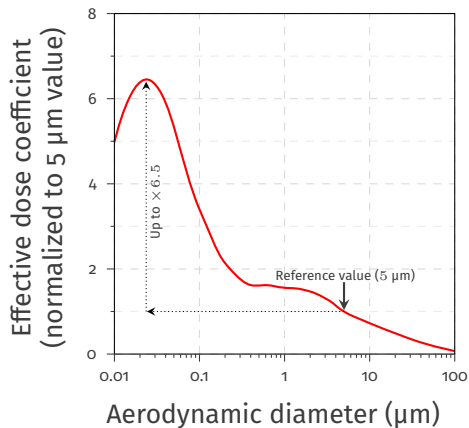
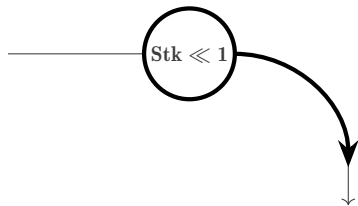
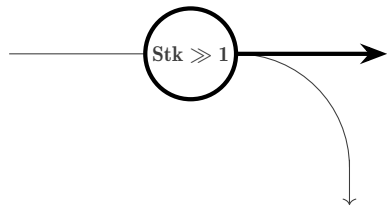


Figure: Influence of particle aerodynamic diameter on the effective dose coefficient (adapted from Laffolley et al., 2026).

Introduction: Inertial separation



A particle with $Stk \ll 1$
follows the streamlines



A particle with $Stk \gg 1$ can't follow the
streamlines due to its inertia

Stokes number represents the probability that a particle will deviate from the streamlines:

$$Stk = \frac{\text{particle response time}}{\text{fluid time scale}} = \frac{\rho_0 C_c(d_{ae}) d_{ae}^2}{18\eta_f} \times \frac{U_f}{D}$$

Introduction: Inertial impactors

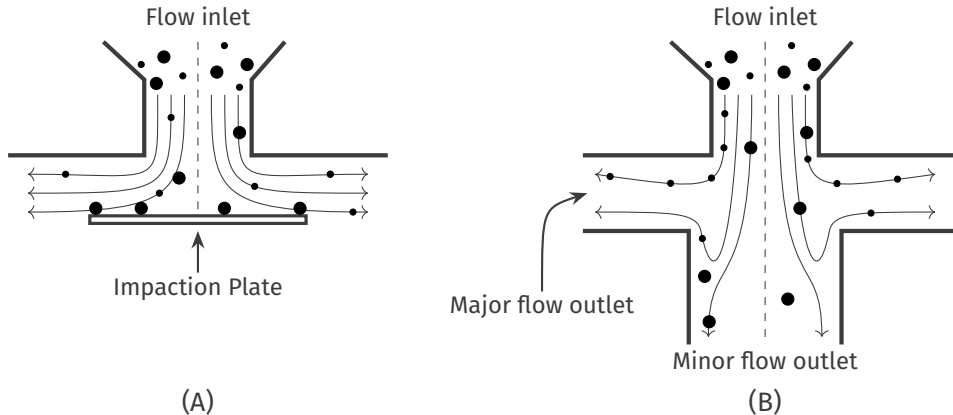


Figure: (A) Conventional plate impactor, (B) Conventional virtual impactor (adapted from Marple et al., 2011).

Introduction: Virtual Impactors' Key Performance Indicators (KPIs)

The **Cutoff diameter** (d_{50}) represents the diameter of the particle that has 50% chances to go in the minor flow outlet. This is the reference size used to separate coarse and fine particle.

$$d_{50} = \sqrt{\frac{18\eta_f D}{\rho_0 C_c(d_{50}) U_f}} \times \sqrt{Stk_{50}}$$

The **Sharpness** (σ) represents the standard deviation of particles around d_{50} .

$$\sigma = \sqrt{\frac{d_{84}}{d_{16}}}$$

(Le et al., 2021)

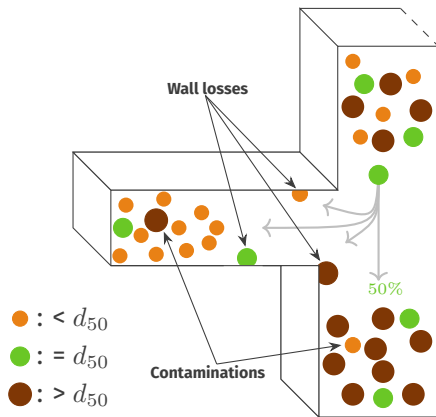


Figure: Repartition of particles in a virtual impactor according to their size

Introduction: μ SPLIT project

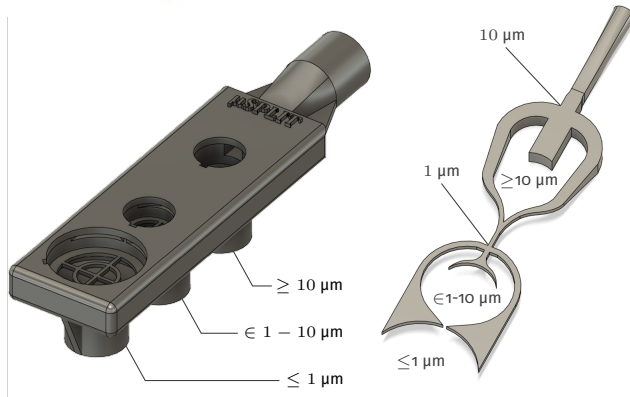


Figure: μ SPLIT project developed by CLADS in JAEA

Introduction: Objectives

CLADS's Requirements :

- ① **Cutoff diameter** target : $d_{50} = 10\ \mu\text{m}$
- ② Elimination of **Major flow contamination**
- ③ Elimination of **Wall losses**
- ④ Elimination of **minor flow contamination**
- ⑤ Reduction of **Sharpness**

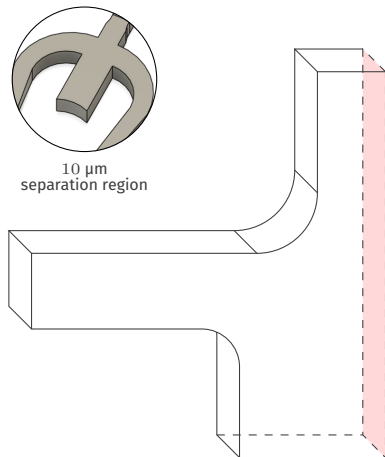
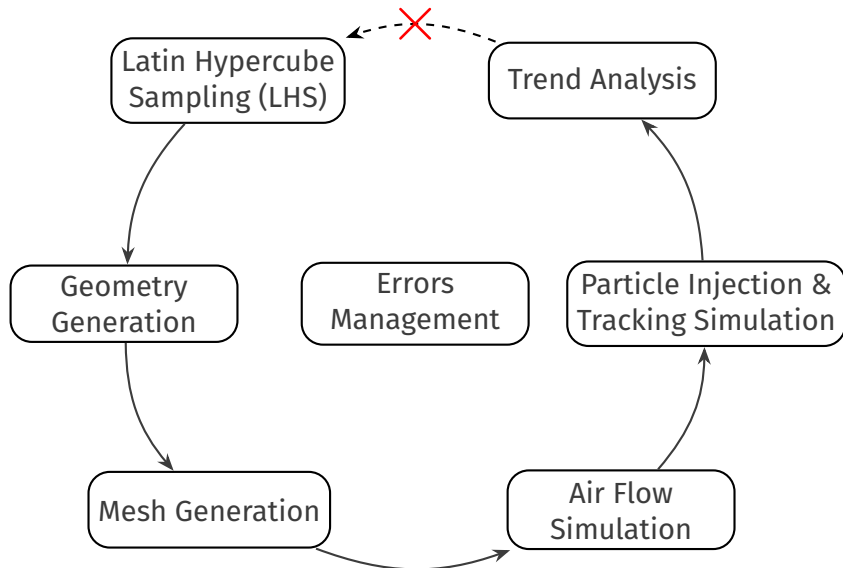


Figure: Half (symmetric) separation region

Methodology: Automation workflow



Methodology: Modeling Assumptions

Individual hypothesis

- Incompressible flow ($Ma < 0.3$)*
- Steady state air flow
- Spherical particles (d_{ae})
- Log-normal distribution
($\sigma = 2.5, \mu = 6.4$)

Coupling hypothesis

- One-way coupling ($\Phi_p < 10^{-6}$)
- Particle-Particle interaction :
No collisions
- Particle-wall interaction :
adhesion (no rebound)
- No slip at walls

Other hypothesis: 3D printing device resolution*, Half geometry consideration

**** $\Rightarrow$ Constraints for LHS Parameters**

Methodology: How to define good LHS Parameters range ?

We must associate :

→ **The 2 Constraints of LHS parameters from previous assumptions :**

$$\left\{ \begin{array}{l} * : W_{\min}, T_{\min} \geq \text{Resolution} \\ * : W_{\min} = \frac{q_{\text{inlet}}}{0.3 \times c_{\text{air}} T_{\min}} \end{array} \right.$$

→ Other settings are mostly issued from literature.

Note : We want to have the wider ranges to explore the parameters as much as possible.

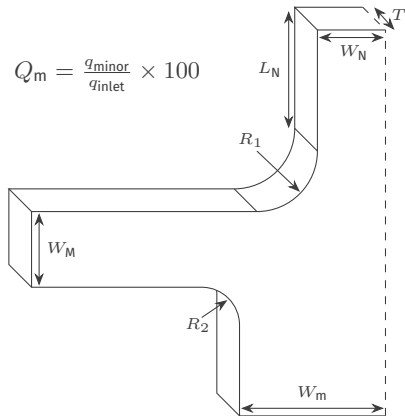


Figure: LHS variable parameters

Methodology: Boundary conditions

	Mass flow rate	Relative pressure	Rebound model
Inlet	$+q_{\text{Major}} + q_{\text{minor}}$	0 Pa / ($\text{kg} \cdot \text{m}^{-3}$)	rebound
Minor outlet	$-q_{\text{minor}}$	Neumann	escape
Major outlet	$-q_{\text{Major}}$	Neumann	escape
Symmetry	sym	sym	sym
Walls	noSlip	Neumann	escape

Table: Boundary conditions

Neumann: $\left. \frac{\partial p}{\partial n} \right|_{n \rightarrow 0} = 0$

noSlip: $U_{\text{wall}} = 0 \text{ m} \cdot \text{s}^{-1}$

sym: Acts as a symmetry plane

escape: Particle sticking model

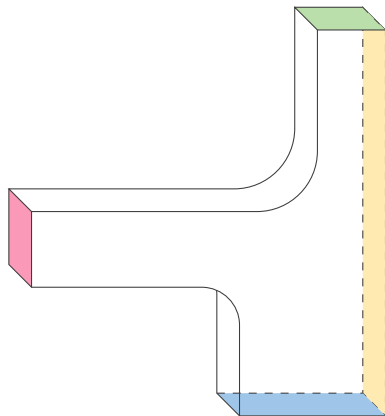


Figure: Identification of boundary regions

Methodology: Numerical Simulation Parameters

CFD parameters

- Steady-state convergence of air flow then particle injection
- Turbulence: RANS $k - \omega$ SST (Separation, recirculations)
- $q_{\text{minor}} = Q_{\text{m}} \times \rho_{\text{air}} \times 5 \text{ L} \cdot \text{min}^{-1}$
- Log-normal distribution ($\sigma = 2.5, \mu = 6.4$)
- Parcel transportation time = $10 \times$ Parcel injection duration

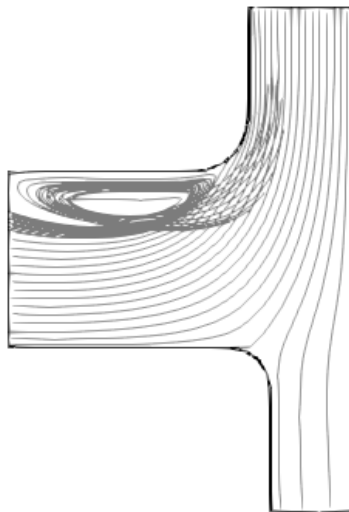
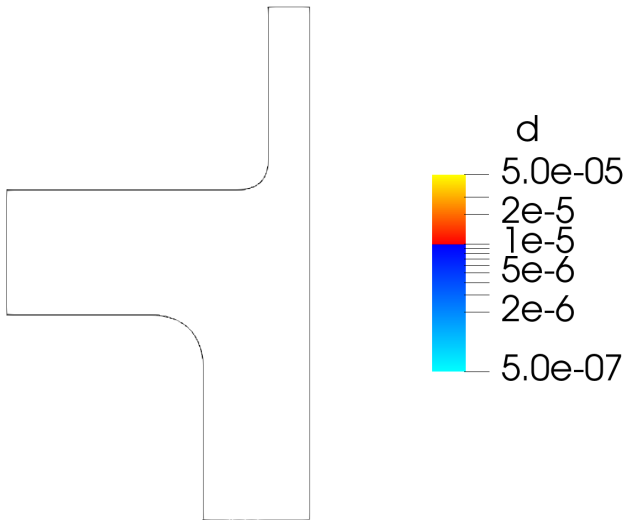


Figure: Streamlines of one geometry

Methodology: Numerical Simulation



Methodology: Mesh and inversed volumes

Meshing issues

- cfMesh tool not adapted to fillets but easily integrable to Python automation.
- snappyHexMesh adapted to any geometries but hard to integrate in the automate.
- Minimal skewness in the fillets is 1.66 (<0.8 recommended).
- Resulting in 36% (18/50) of converged airflow simulations.

(Despite the mesh independence study carried out beforehand)

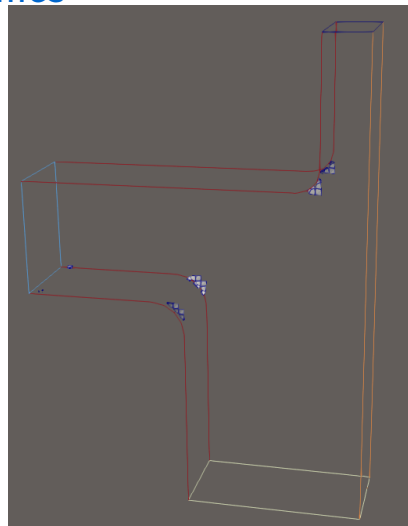


Figure: Inversed volumes created by cfMesh in the fillets

Methodology: Design of Experiments

“Solution” of this problem: More simulations

- For Loeppky et al. (2009) good exploration relies on 10 experiences per parameter (80 converged simulation needed for 8 parameters).
- Need 250 ($\approx \lceil 10 \times 8 / 0.36 \rceil \times 1.1$) configurations to obtain a minimum of 80 converged simulation.
- Utilization of HPC + Decomposition in subregions for parallel computing.

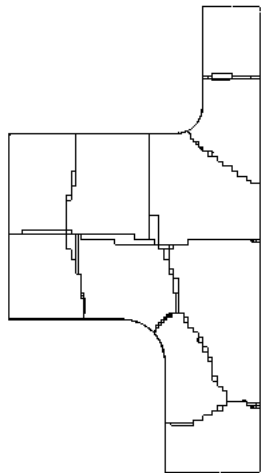
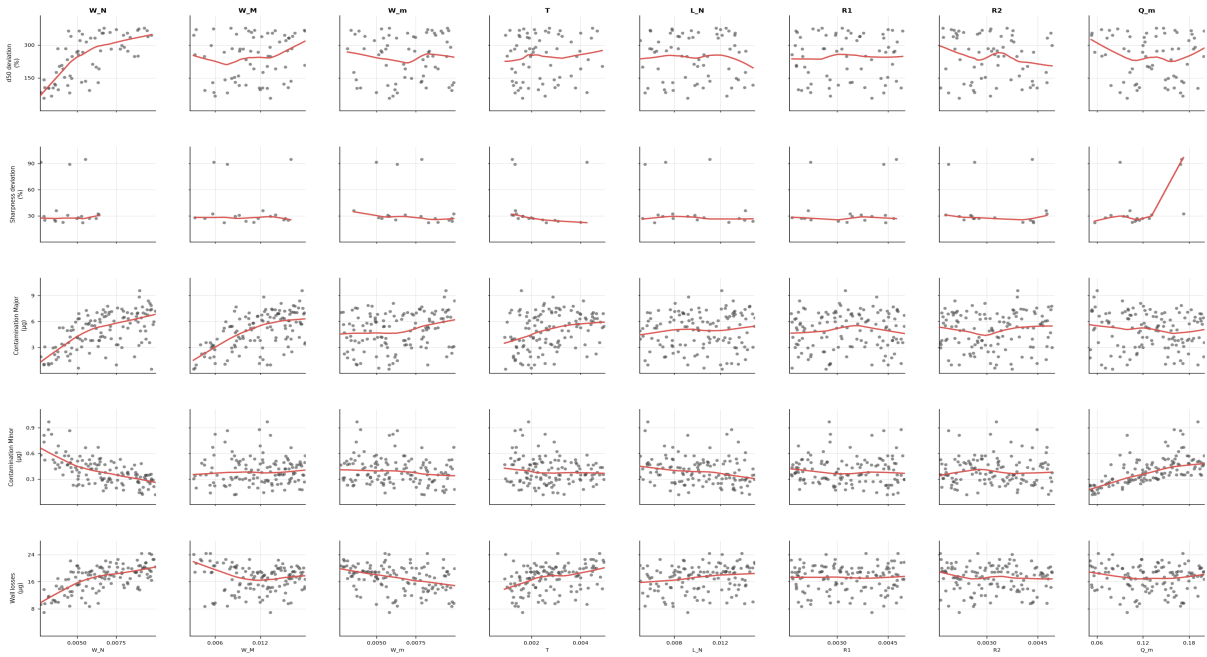
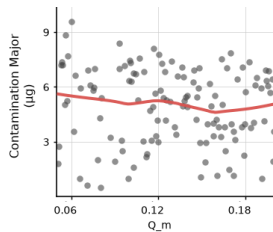
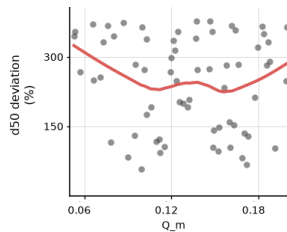


Figure: Regions of a Decomposed Case (scotch method)

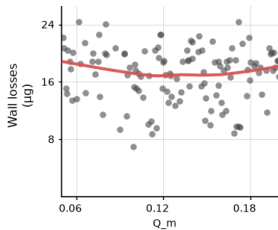
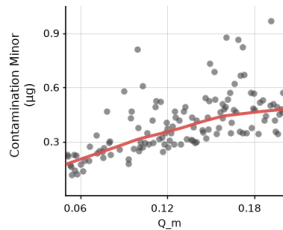
LHS Synthesis — seed 01 — 125/250 cases



Results: How to determine optimal configuration - Example with Q_m



$$Q_m \in 5 - 20 \%$$



- ① **Cutoff diameter** target : $d_{50} = 10 \mu\text{m}$
- ② Elimination of **Major flow contamination**
- ③ Elimination of **Wall losses**
- ④ Elimination of **minor flow contamination**

Results: Optimal configuration's separation performance

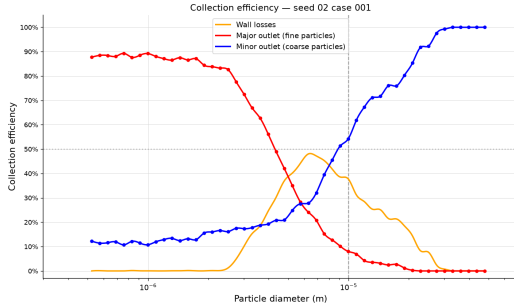


Figure: Collection efficiency diagram with wall losses.

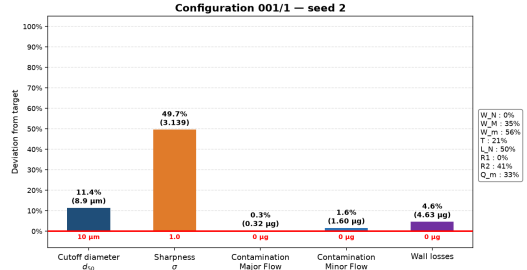


Figure: Deviation of this optimal configuration's KPIs to ideal ideal configuration of virtual impactor.

Results: Radioprotection considerations

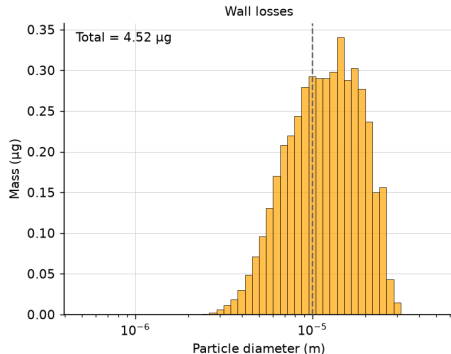


Figure: Mass collection on the walls of this configuration

- **4.52 μg** of particles deposited on the walls in 1 operating cycle,
- **1 %** of ^{137}Cs in the total particles deposited and $e(50)$ issued from ICRP (1995),
- **Entire inhalation** of the deposited sample by the operator.

$$D_{\text{eff}} = a_s \times m_{^{137}\text{Cs}} \times e(50)$$
$$\Rightarrow D_{\text{eff}} \approx \mathbf{0.665 \text{ mSv/operating cycle}}$$

Maximal effective dose (ICRP, 2007):

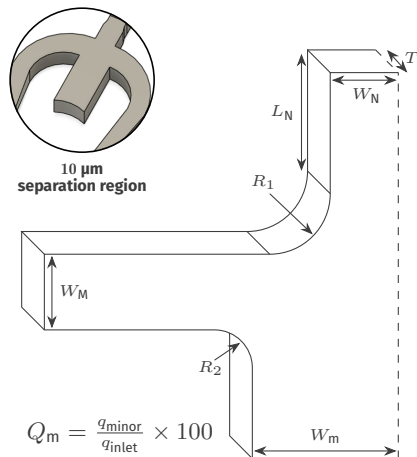
- Receivable by a civilian: 1 mSv/year
- Receivable by an operator: 20 mSv/year

$\Rightarrow$ **Single use 3D printed μSPLIT is a necessity.**

Conclusion

Internship Achievements :

- **Automation & Parametric study**
LHS, CAD, CFD, Postprocessing
- **Optimal Configuration's Performance**
 - 11.4% (8.9 μm) deviation from CLADS's objective
 - Low contamination at both outlets
- **μSPLIT Project**
 - $D_{\text{eff}} \approx 0.665$ mSv for 1 operating cycle
 - 3D printed, single use obligation



Thank you Yokoyama-sensei and
Hugo-san!

横山先生、ヒューゴさん、本当にあ
りがとうございました。

Thank you for your attention
ご清聴ありがとうございました。

Appendices

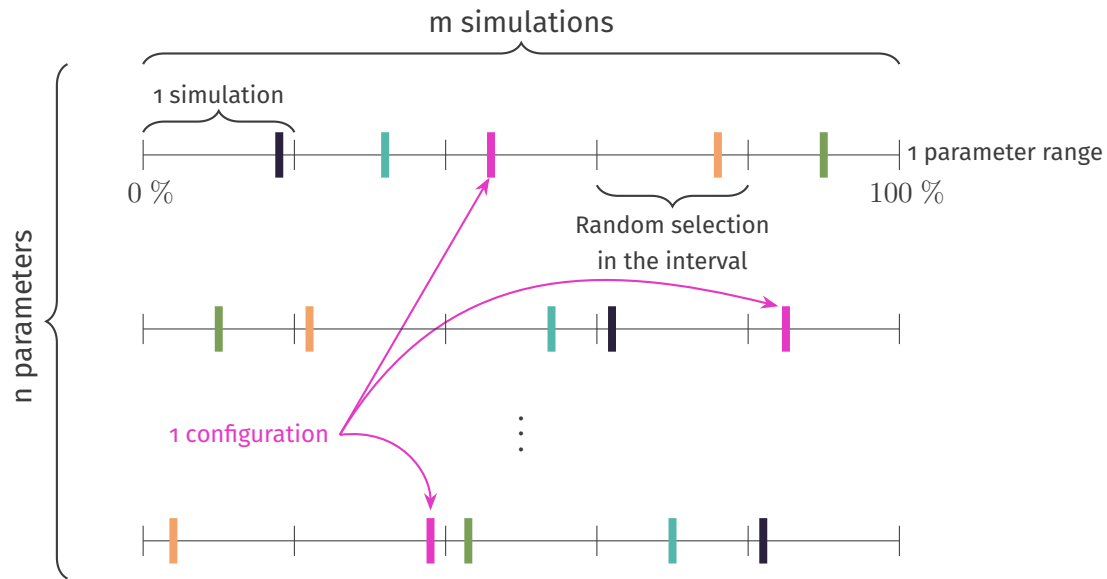
Contacts

Ryo YOKOYAMA - Internship Academic Supervisor
Research Associate at University of Tokyo (UTokyo)
yokoyamaryo395@g.ecc.u-tokyo.ac.jp

Hugo LAFFOLLEY - Internship Supervisor
Former JAEA Postdoctorant - Current EDF Engineer

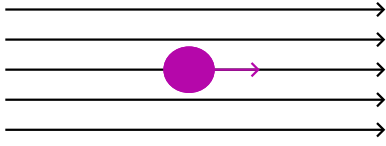
Youichi TSUBOTA - Responsible of the μ SPLIT project
Manager in the Japanese Atomic Energy Agency (JAEA)
tsubota.yoichi@jaea.go.jp

Automation: Latin Hypercube Sampling (LHS)

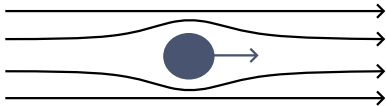


Coupling regime

One-way



Two-way



Four-way

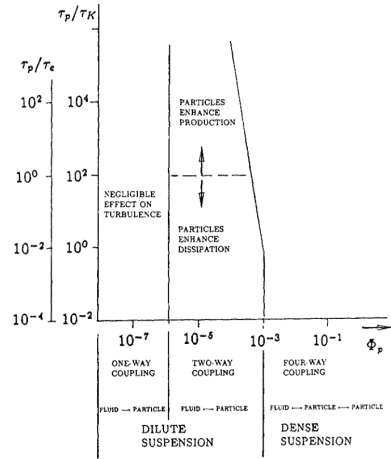
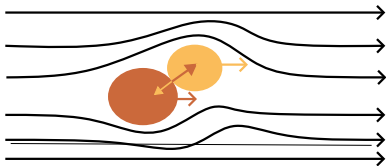
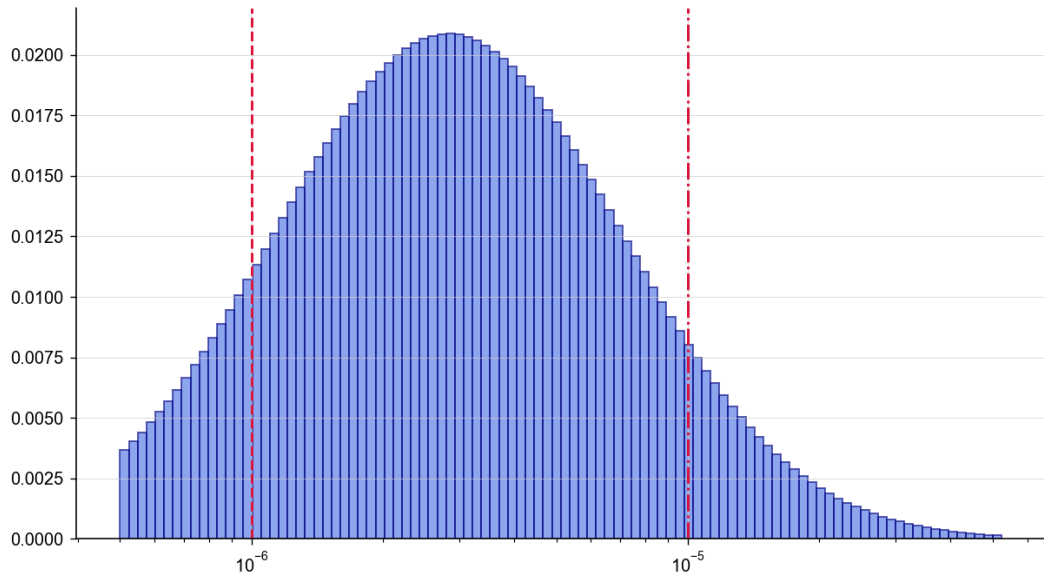
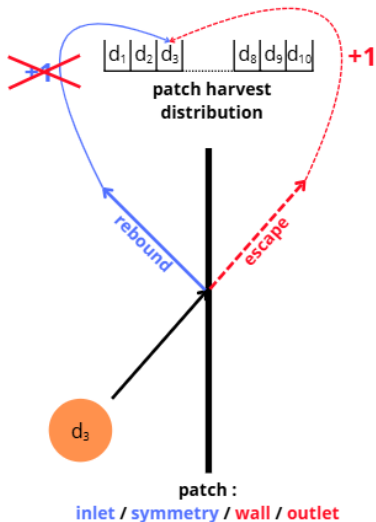


Figure: Map of regimes of interaction between particles and turbulence (Elghobashi, 1994).

Log-normal distribution ($\sigma = 2.5$, $\mu = 6.4$) (Zhuang et al., 2026)



Rebound model



$$e = \left(1 - \left(\frac{v_s^n}{v_i^n} \right)^2 \right)^{1/2}$$

→ $v_s^n = f(d_p, \text{Physical properties of surface})$

→ v_i^n issued from CFD (Thornton et al., 1998)

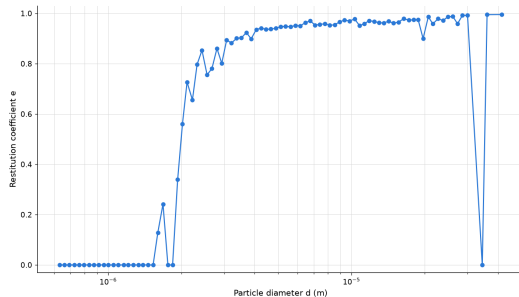
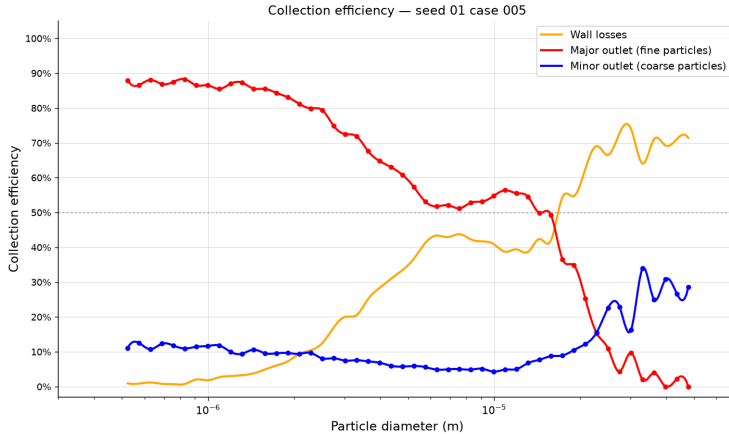


Figure: Coefficient of restitution e of the kinetic energy of the particle in function of its diameter d_p .

Sharpness issue explanation

$$\sigma = \sqrt{\frac{d_{84}}{d_{16}}}$$



Experimental validation

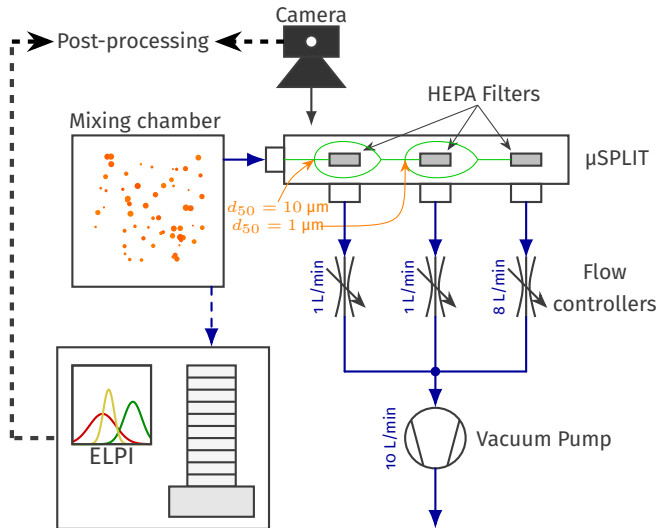


Figure: Experimental setup

3D Printing considerations

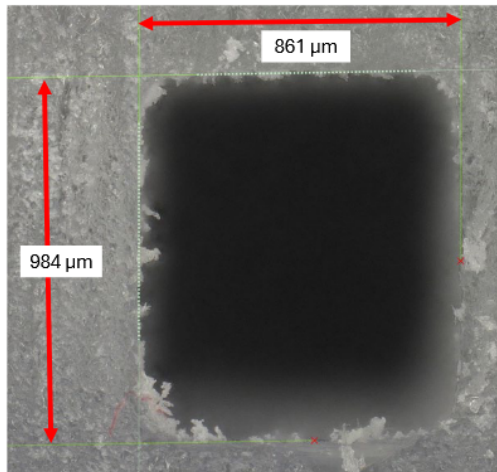
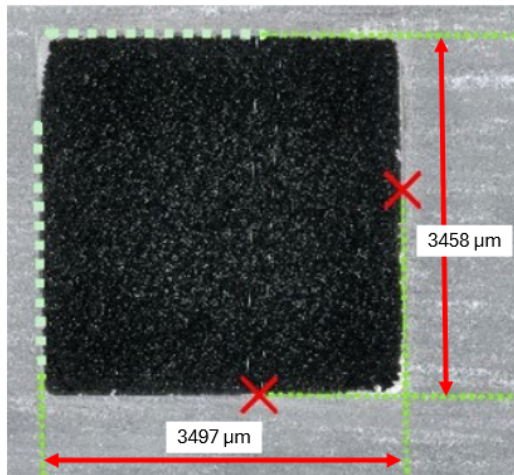
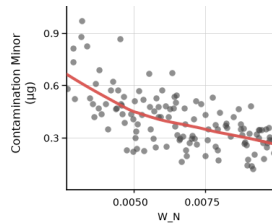
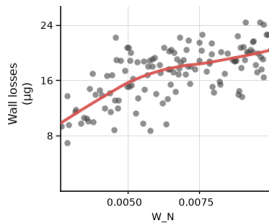
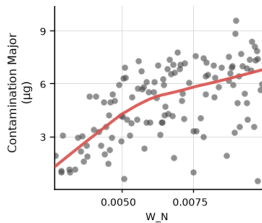
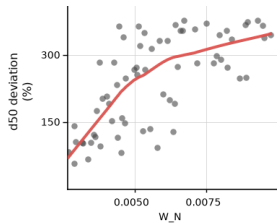
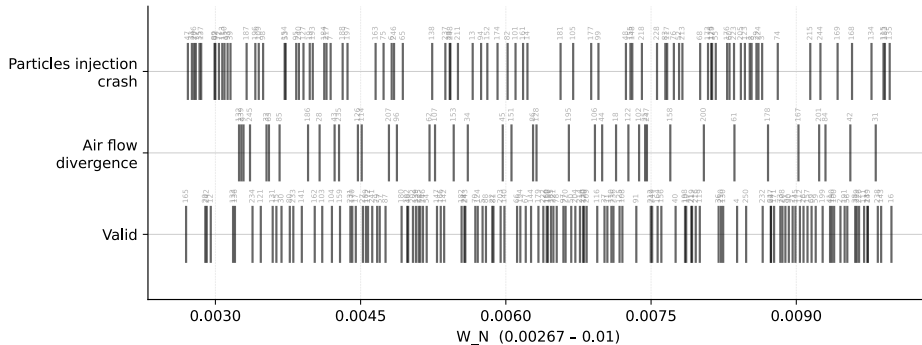
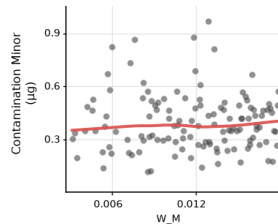
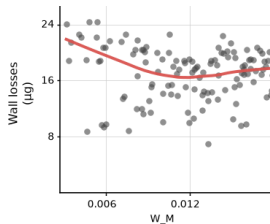
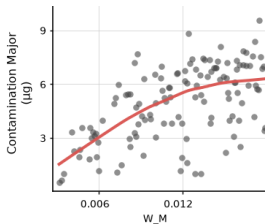
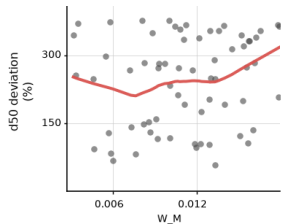
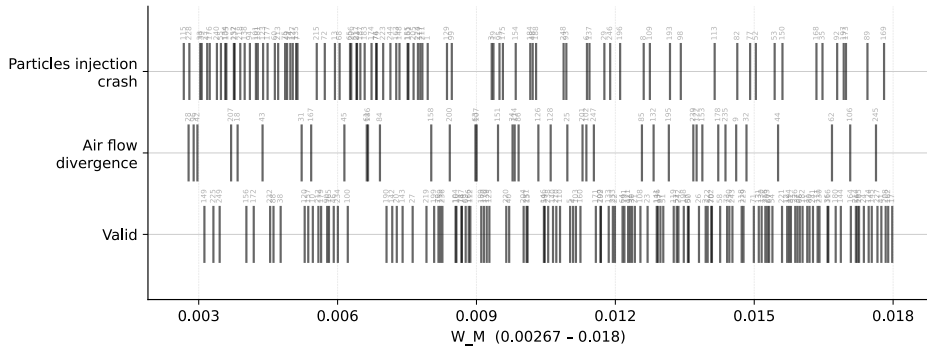
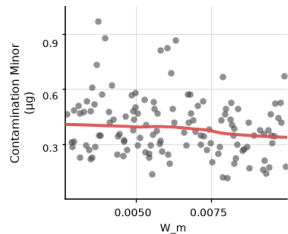
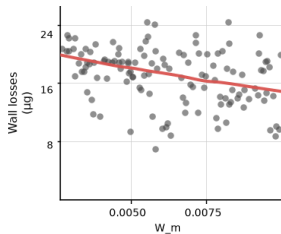
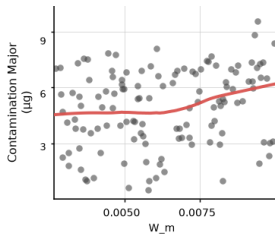
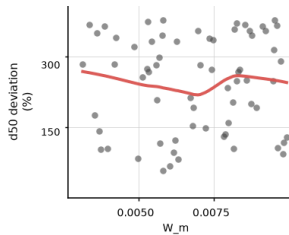
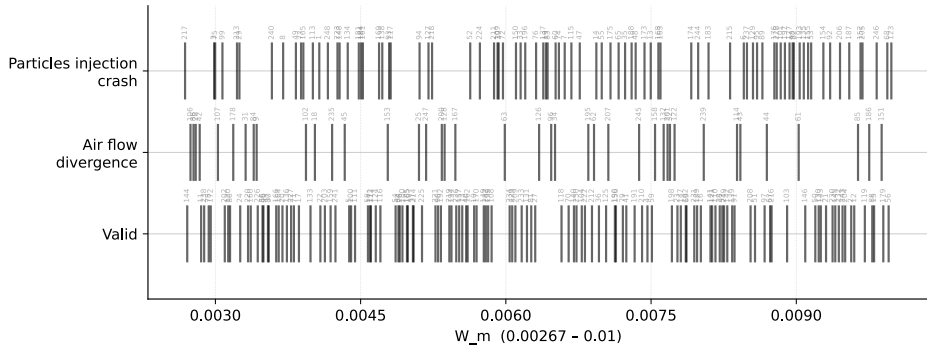
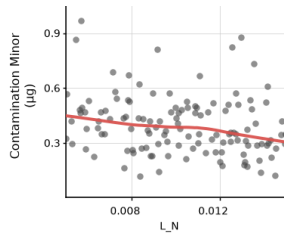
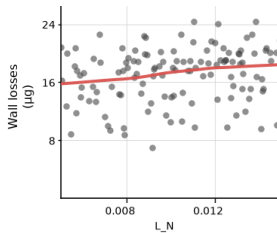
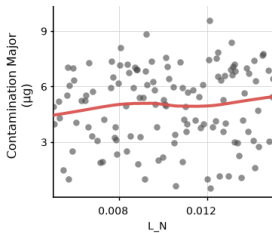
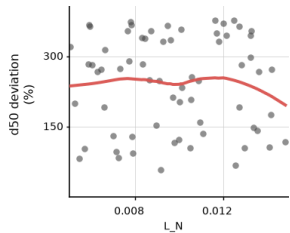
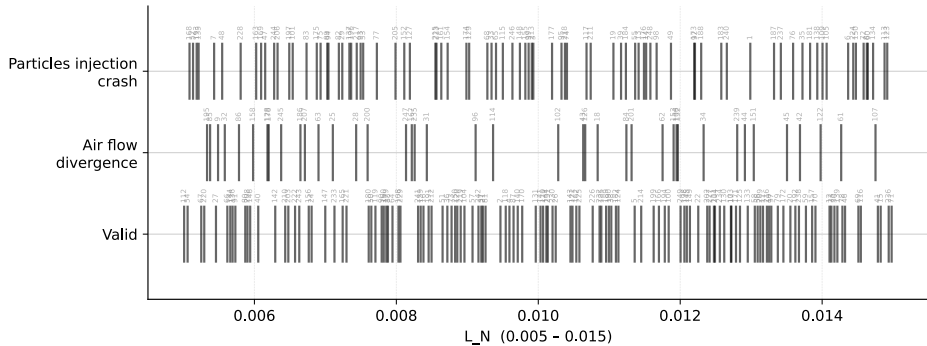


Figure: Printing defaults on the 10 μm stage (left) and on the 1 μm stage (right) (Laffolley et al., 2026).

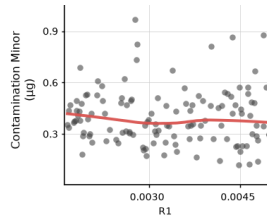
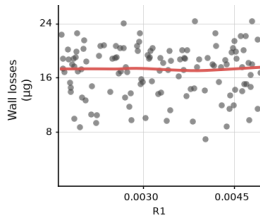
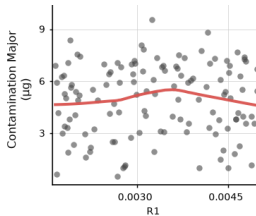
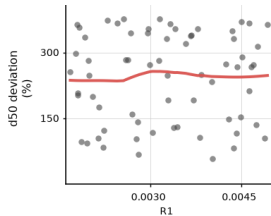
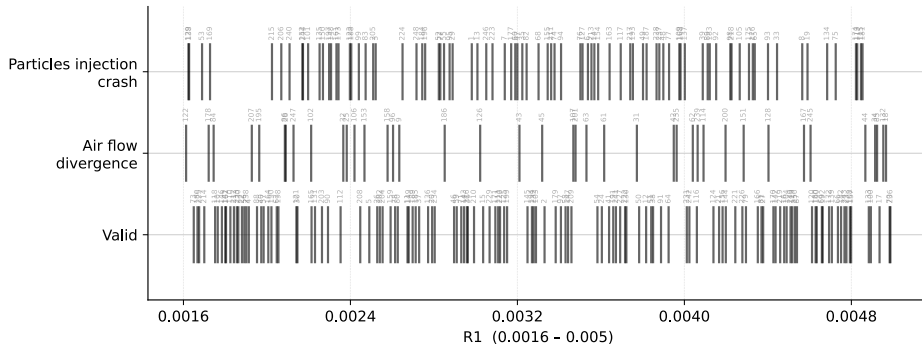


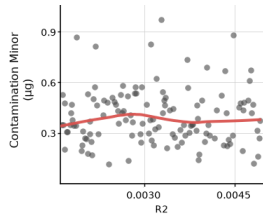
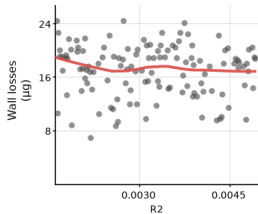
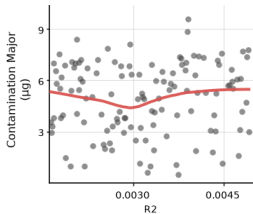
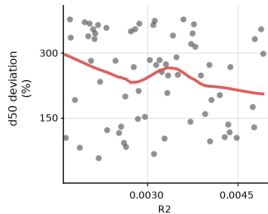
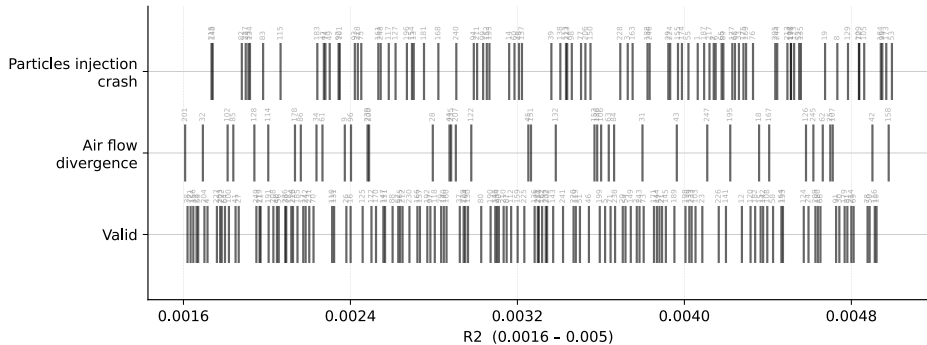




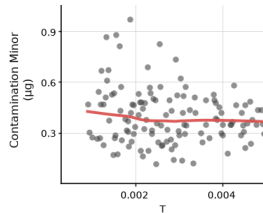
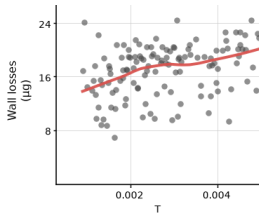
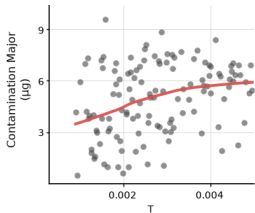
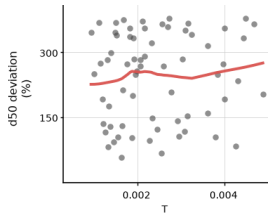
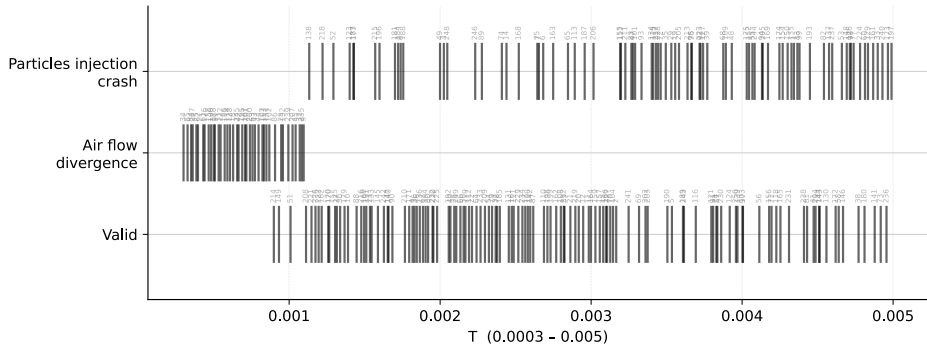
L_N 

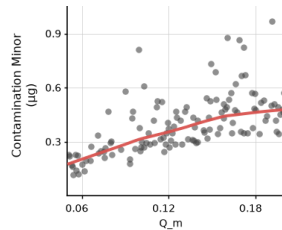
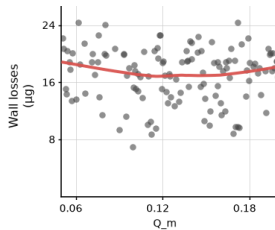
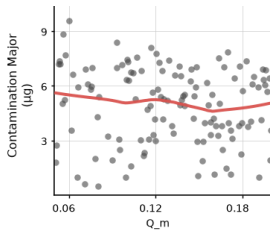
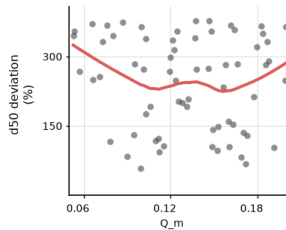
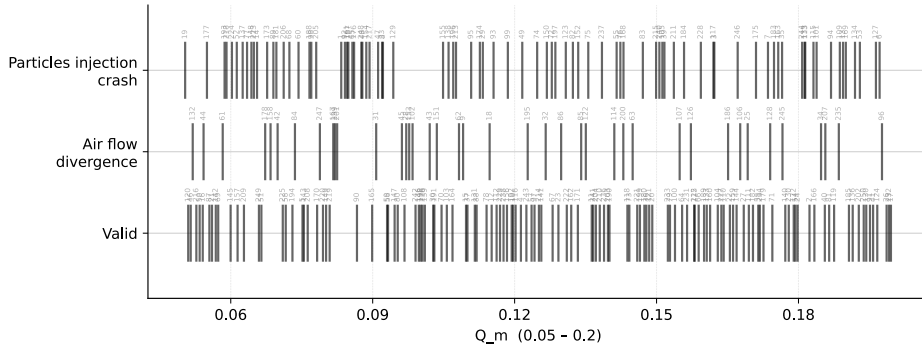
R_1



R_2 

T





References

- Colbeck, Lazaridis (2014) “Aerosol science: Technology and applications” John Wiley & Sons
- Laffolley, Tsubota (2026) “Development of a low-cost and disposable virtual impactor for particle-size-dependent radioactivity assessment at Fukushima Daiichi NPS”. In: *AESJ 2026 Annual Meeting*
- Marple, Olson (2011) “History of Virtual Impactors”. In: *Aerosol science and technology: history and reviews* p.509
- Le, Tsai (2021) “Inertial Impaction Technique for the Classification of Particulate Matters and Nanoparticles: A Review”. In: *KONA Powder and Particle Journal* vol.38
- Loeppky, Sacks et al. (2009) “Choosing the sample size of a computer experiment: A practical guide”. In: *Technometrics* Vol. 51, p. 366–376 Taylor & Francis

References

- ICRP (1995) “Age-dependent Doses to Members of the Public from Intake of Radionuclides - Part 5 Compilation of Ingestion and Inhalation Coefficients”. In: *ICRP Publication 72*
- ICRP (2007) “Recommendations of the International Commission on Radiological Protection”. In: *ICRP Publication 103*
- Elghobashi (1994) “On predicting particle-laden turbulent flows”. In: *Applied scientific research* Vol. 52, p. 309–329 Springer
- Zhuang, Fang et al. (2026) “Wet deposition modeling and behavior analysis of ^{137}Cs aerosols using ten observation-based continuous size distributions following the Fukushima accident”. In: *Environmental Pollution* Elsevier
- Thornton, Ning (1998) “A theoretical model for the stick/bounce behaviour of adhesive, elastic-plastic spheres”. In: *Powder technology* Vol. 99, p. 154–162 Elsevier